Retroviruses and Human immunodeficiency virus (HIV) and acquired immunodeficiency syndrome (AIDS)

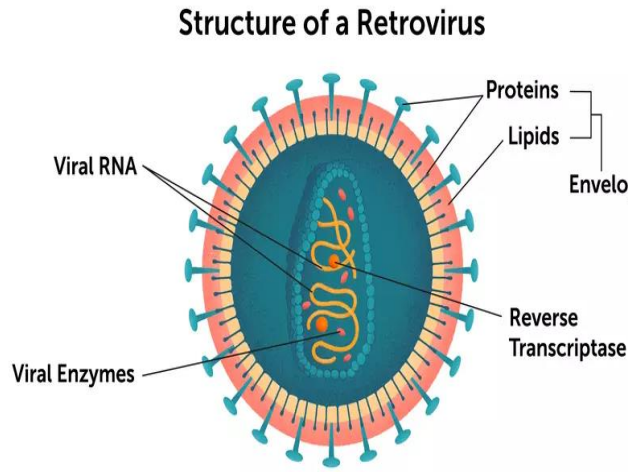
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INTRODUCTION



Retroviruses :

- The prefix retro means “backward,”
- Retrovirus refers to the fact that these viruses transfer information from RNA to DNA .
- Retroviruses are structurally complex animal viruses that contain an RNA genome .
- They are spherical, enveloped and carry several enzymes within the virion (reverse transcriptase, integrase,and a retroviral-specific protease)
- Their genome consists of two identical positive sense ssRNAs

Classification

They classified into three subfamilies

Oncovirinae (HTLV-1, HTLV-2, HTLV-5)
Lentivirinae (HIV-1, HIV-2)
Spumavirinae

Retrovirus species of medical and veterinary importance include:

i. Human Immunodeficiency Virus (HIV-1 and HIV-2), the causative agents of Acquired Immunodeficiency Syndrome (AIDS);
ii. Human T-cell Leukemia Virus Type 1 (HTLV-1), associated with adult T-cell leukemia/lymphoma (ATLL) and tropical spastic paraparesis;

Human immunodeficiency virus (HIV)

- Etiologic agent of Acquired Immunodeficiency Syndrome (AIDS).
- Discovered independently by Luc Montagnier of France and Robert Gallo of the US in 1983-84.
- Former names of the virus include:

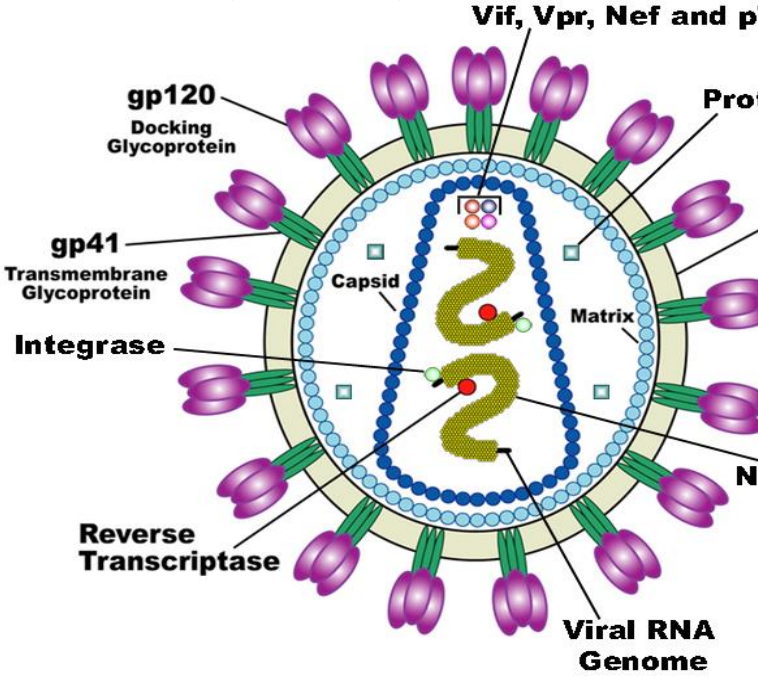
Human T cell lymphtrophic virus (HTLV-III) , Lymphadenopathy associated virus (LAV) & AIDS associated retrovirus (ARV)

- HIV-1 is the most well-studied retrovirus, affecting over 38 million people worldwide and remaining a major global public health challenge despite advances in antiretroviral therapy.

Human immunodeficiency virus (HIV)

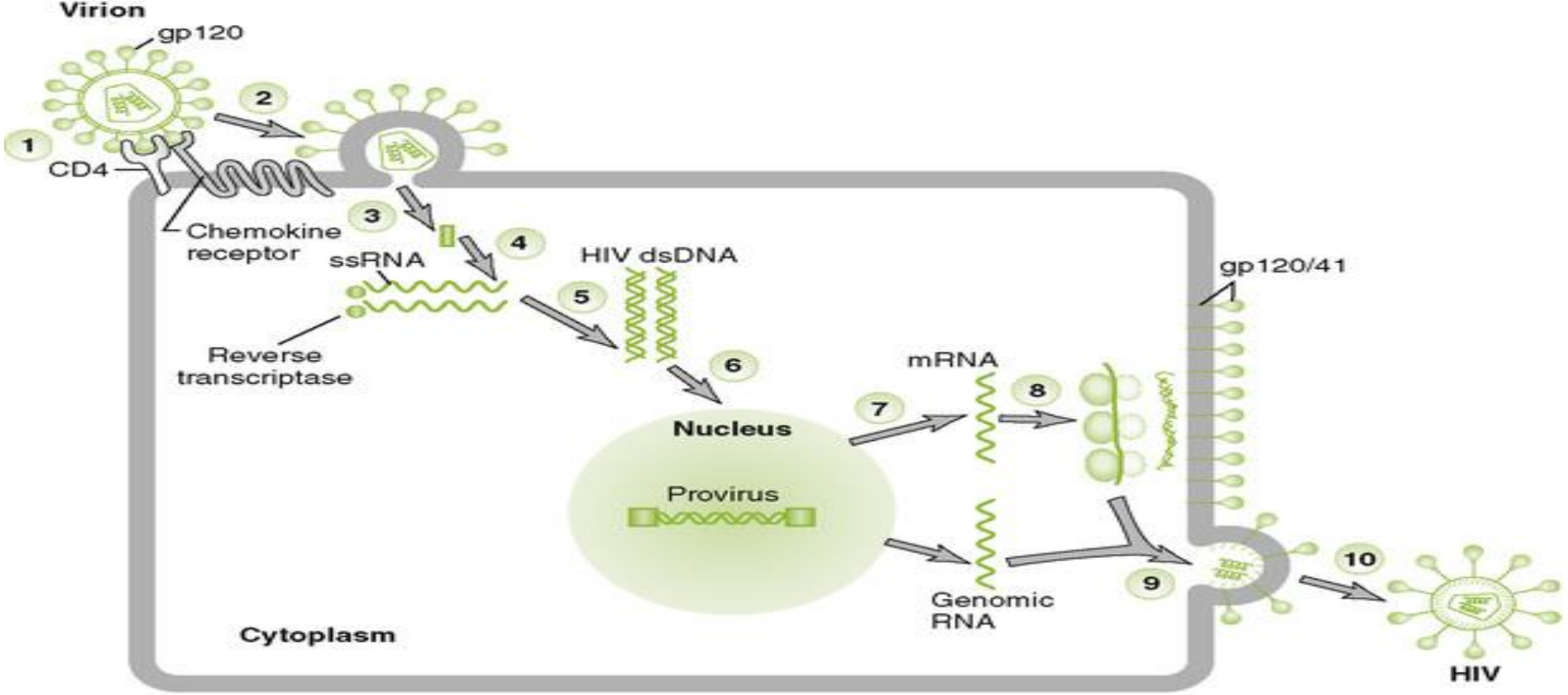
- RNA virus
- Genus: Lentivirus
- Family: Retroviridae(retro-reverse).
- HIV viruses:

  HIV 1 & HIV2



Characteristics of the virus

- Retroviruses transcribe RNA to DNA.
- Two viral strands of RNA found in core surrounded by protein outer coat.
- Outer envelope contains a lipid matrix within which specific viral glycoproteins are imbedded .
- These knob-like structures responsible for binding to target cell.



1. Binding of HIV gp120 to CD4 and chemokine coreceptor (early disease: CCR5; late disease: CXCR4)
2. Fusion of virion with cell membrane mediated by HIV gp41 and a coreceptor on cells
3. Release of nucleocapsid into cytoplasm
4. Uncoating of nucleocapsid and release of 2 copies of single-stranded (ss) RNA genome and reverse transcriptase
5. Formation of HIV double-stranded (ds) DNA by viral reverse transcriptase
6. Transport of HIV dsDNA into nucleus and its integration into host chromosomal DNA by viral integrase forming provirus
7. Transcription of proviral DNA into genomic RNA and several mRNAs, which are transported to cytoplasm
8. Synthesis, processing, and assembly of viral proteins in or near cell membrane
9. Movement of HIV genomic RNA to modified membrane and budding out to form envelope
10. Release of virion by budding off and subsequent cleavage of polyproteins by viral protease to yield mature virion

HIV Structure.

The core Enzyme needed for the development of the virion such as reverse transcriptase, protease, double nuclease and intergrase are also present.

- Two positive sense viral RNA
- P24 Caspid protein that forms the protective shell around viral genome and enzym

The Matrix

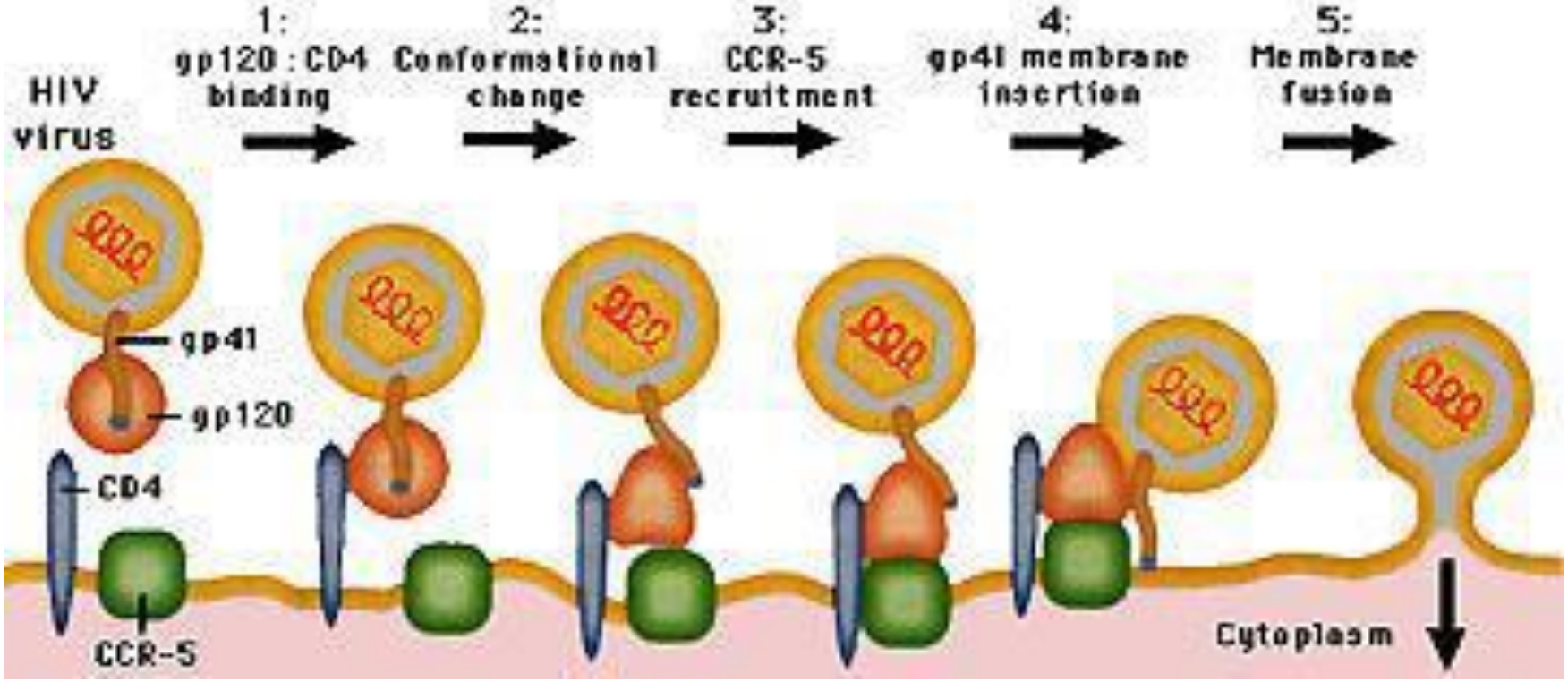
- composed of viral proteins p17 surrounds the capsid ensuring the integrity of the virion particle.
- It also helps in directing the virus to the nucleus after infection

The Envelope

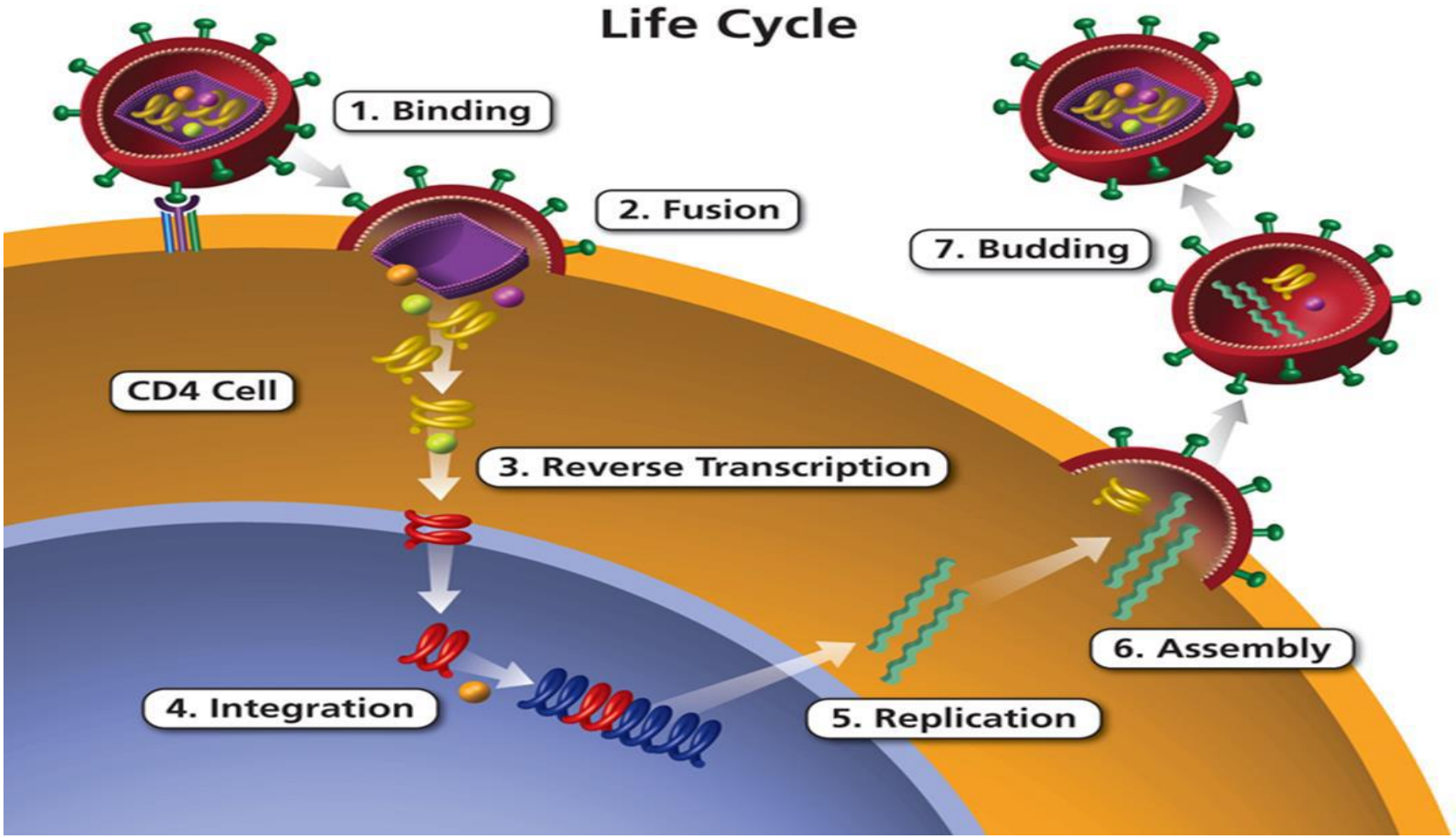
- It is composed of glycoproteins like gp41 and gp120 Which are essential for viral entry into the host cell by binding to CD4 receptors CCR5 and CXCR4
- It also consist of lipid bilayer derived from host cell membrane

Viral Replication:

- Viral Entry: HIV entry requires the presence of CD4 + ,one of two co---receptors: CCR5 or CXCR4 .
- The gp120 protein on virus binds specifically to CD4 receptor on host cell with high affinity. Gp41 causes fusion of the virus to the cell membrane.

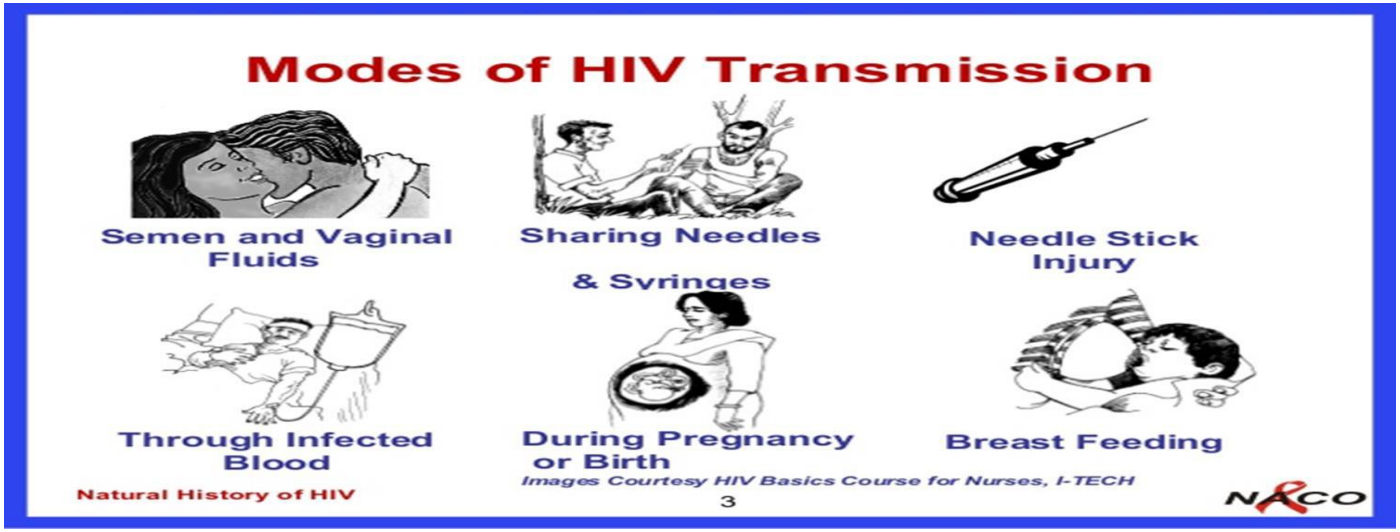


After fusion virus particle enters cell . Viral genome exposed by uncoating particle. Using a viral enzyme called integrase, the viral DNA is integrated into the cellular DNA. Viral RNA is synthesized by the cellular enzyme RNA polymerase .



Methods of transmission:

- Sexual transmission, presence of STD increases likelihood of transmission.
- Exposure to infected blood or blood products.
- Use of contaminated clotting factors by hemophiliacs.
- Sharing contaminated needles (IV drug users).
- Transplantation of infected tissues or organs.
- Mother to fetus.
- HIV is *NOT* transmitted by:
- casual contact ,touching, hugging, (kissing), coughing, sneezing, insect bites, water, food, toilets, and swimming pools or public baths.



Pathogenesis :

HIV destroys CD4+ cells in 3 ways:

1.Viral replication leaves holes in cell membranes .

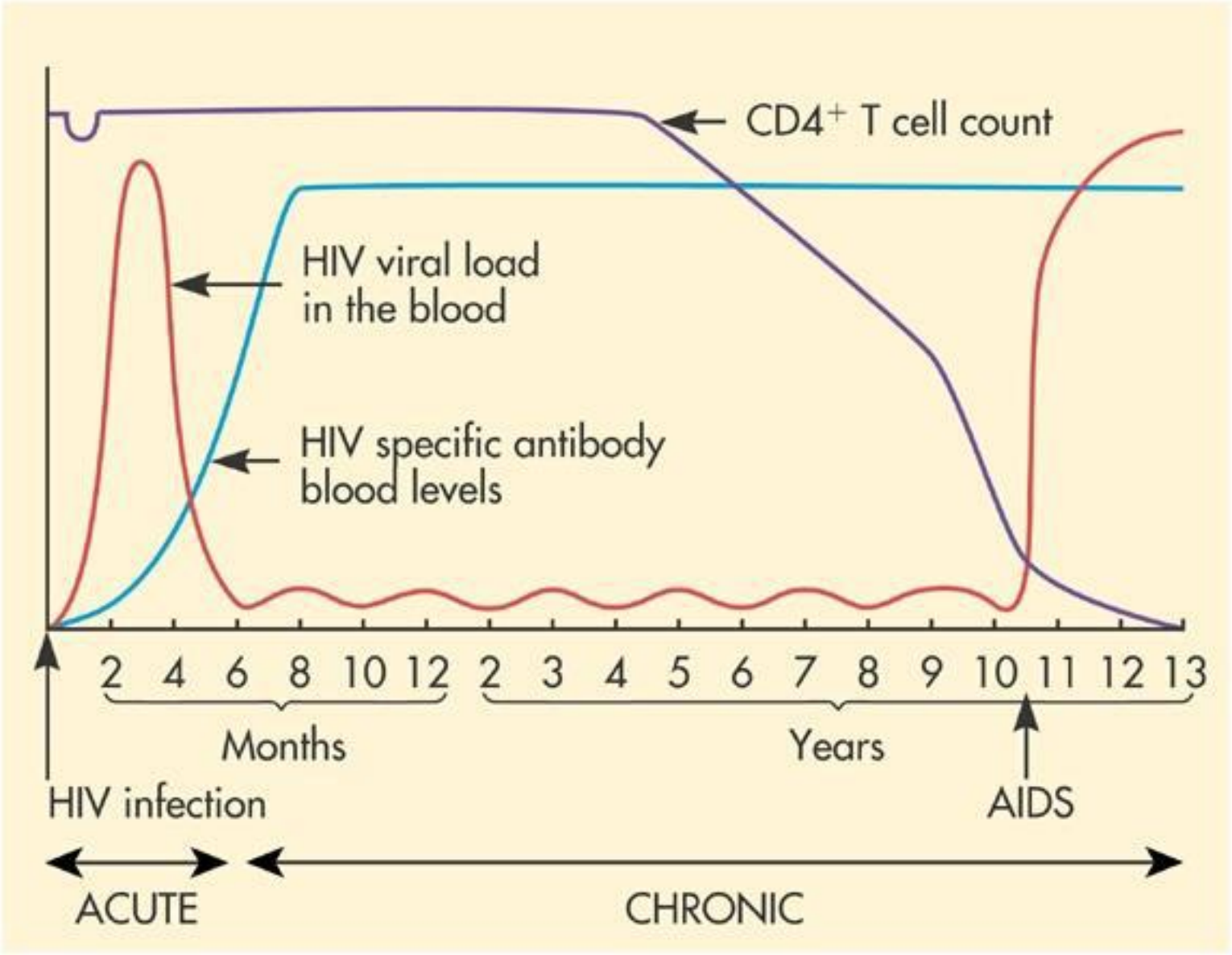
2.Infected cells fuse with other cells, Combine to form a syncytium that destroys all affected cells.

3. Antibodies against HIV bind to the infected cells and activate the complement system, which destroy the infected cells .

- In the immediate period following exposure, HIV is present at a high level in the blood (as detected by HIV Antigen and HIV-RNA assays).
- -It then settles down to a certain low level (set-point) during the incubation period.
- During the incubation period, there is a massive turnover of CD4 cells, whereby CD4 cells killed by HIV are replaced efficiently.
- Eventually, the immune system succumbs and AIDS develop when killed CD4 cells can no longer be replaced (high HIV-RNA, HIV-antigen, and low CD4 counts).

Infectious period :

- Patients are infectious throughout life, &highest during the acute illness phase and *full-blown*AIDS develops.
- **Clinical features :**
- Both HIV 1 and 2 cause clinically indistinguishable illness
- **-Acute HIV illness :**
- Most patients do not have any clinical illness. However, glandular fever-like illness, hepatitis, meningitic illness and thrombocytopenia have been described .There is a high level of HIV viremia at this time.
- **-Asymptomatic chronic infection:**
- After acute infection a steady state evolves between the virus and the immune system. The infected CD4 cells that die are replaced rapidly. This stage may last for 5–15 years and is characterized with low viral load and a normal CD4 lymphocyte count. Some patients may develop persistent generalized lymphadenopathy (PGL).
- **AIDS :**
- This is characterized by an increase in HIV viral load and a decrease in the CD4 lymphocytes, which leaves the patient open to opportunistic infections and other conditions associated with immune deficiency states.



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Laboratory Diagnosis:

1. ELISA

Initial screening, two different ELISA results must be positive before a confirmatory test is performed .

2. Western blot Analysis

Confirmatory test

3. RT-PCR: Detection of virus RNA in blood

4. CD4:CD8: (T-cell ratio)Staging the disease

Negative Antibody Test Results

- HIV negative
- Recent infection: too early for seroconversion
- CDC: follow-up testing at 6 weeks, 12 weeks, 6 months

Significance of ELISA:

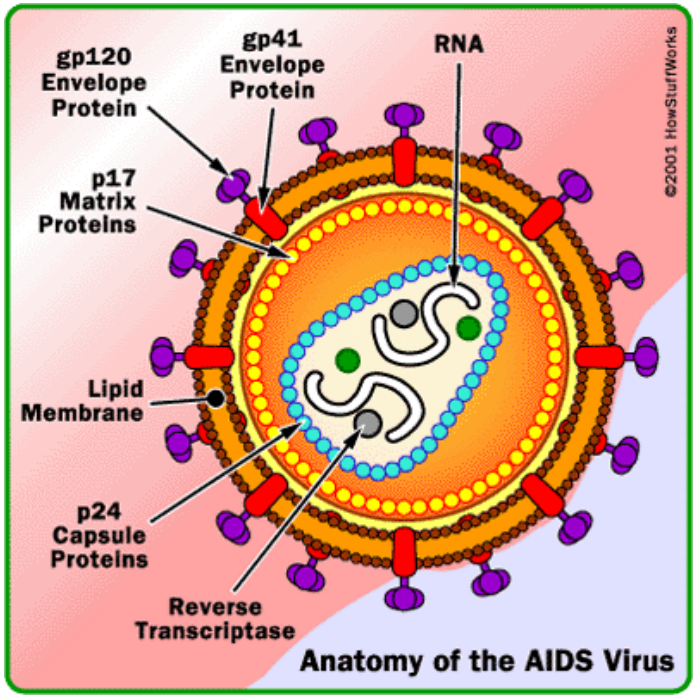
- Most reliable screening test for HIV infection.
- 50% of the infected will show positive reactivity in < 22 days.
  -95% show reactivity < 6 weeks
- Sensitivity > 99.9 %
- But needs confirmation with Western Blot.
- **Determination of P 24 Antigen by ELISA:-**
  Helps in the early detection of HIV infection.
- Screening of Blood for HIV infection along with detection for antibody detection.

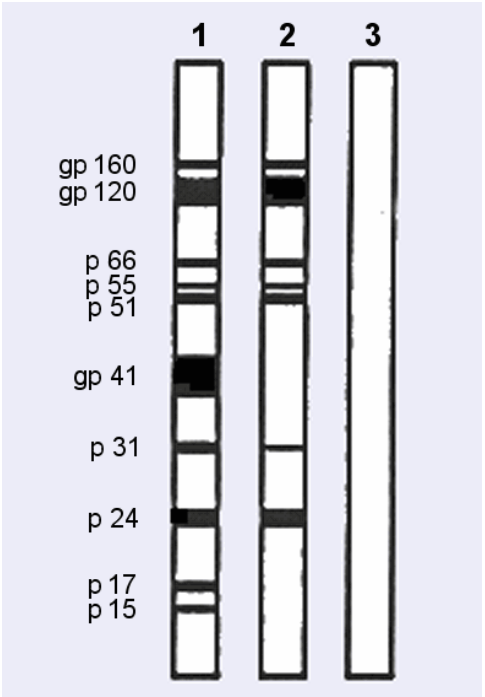
WESTEN BLOT TEST:

- A Gold standard confirmatory test.
- When tested combined with ELISA specificity is > 99.9 %.

Principle of Western Blot :

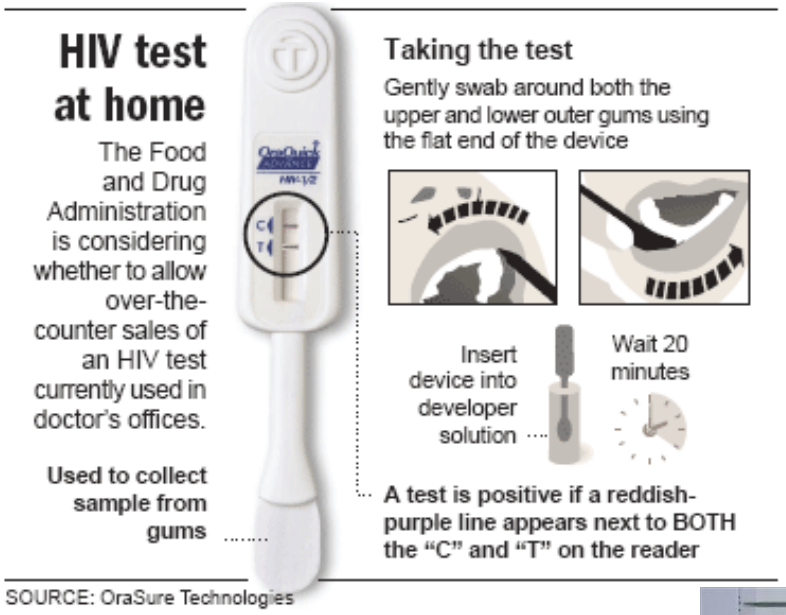
- HIV viral antigens are separated as;
- gp160,gp120,p66,p55,p51,gp41,p31,p24, p17, and p15.
- Interpretation depends on presence / absence of reactivity to specific antigens.

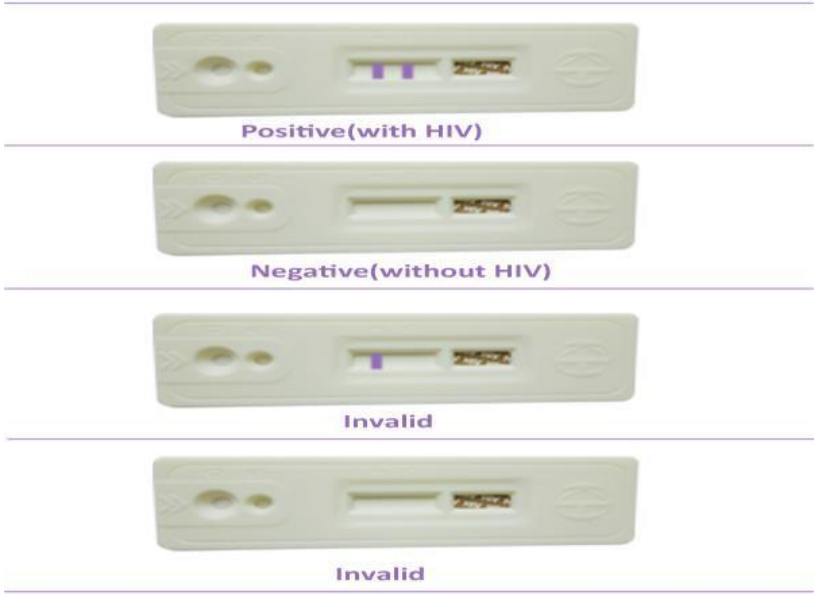




Rapid tests in the Diagnosis of AIDS :

- A growing importance
- Results can be issued within < 20 minutes.
- less demanding technical skills.
- But needs confirmation with ELISA / Western Blot testing.

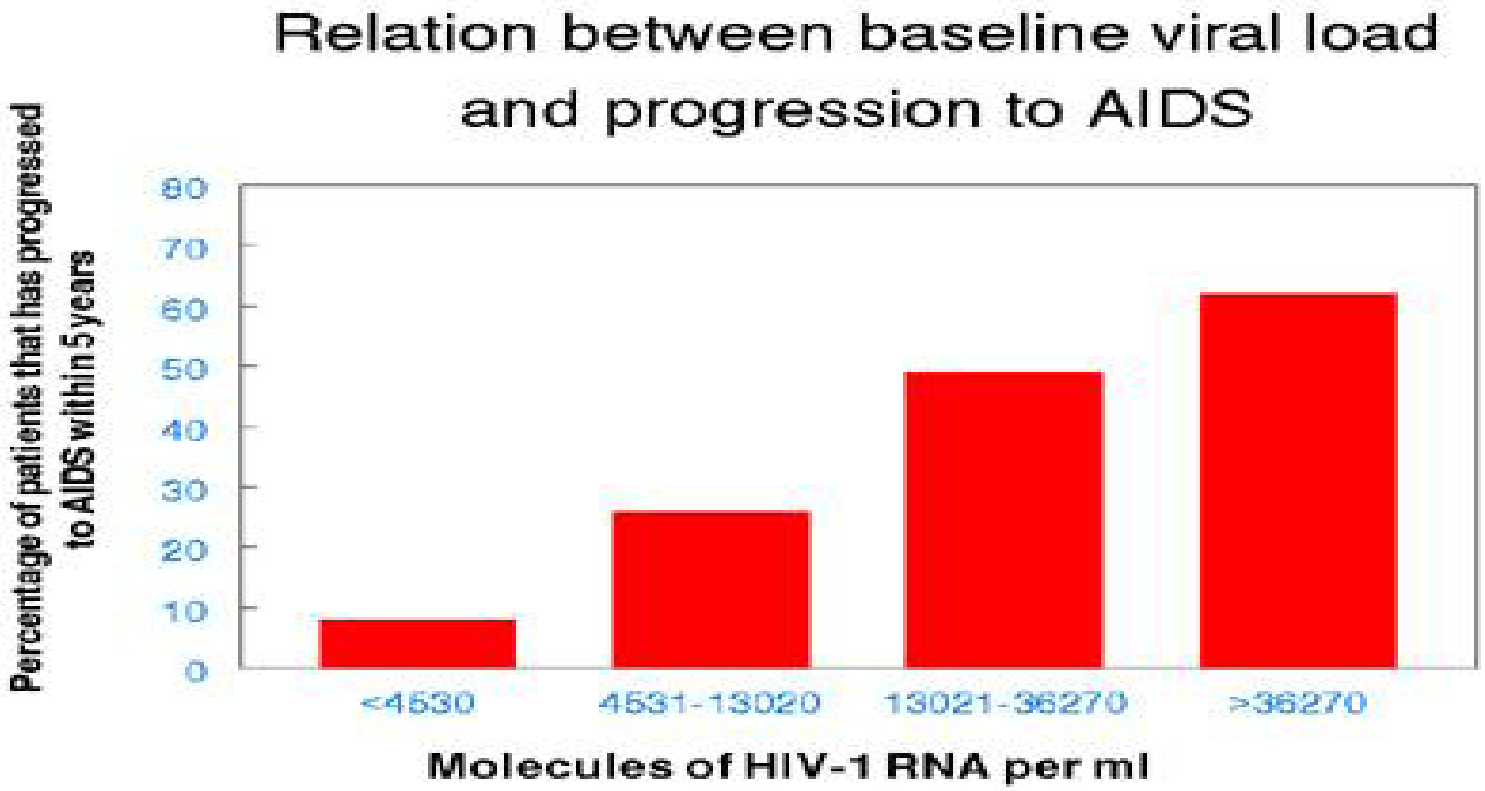






HIV Viral Load tests :

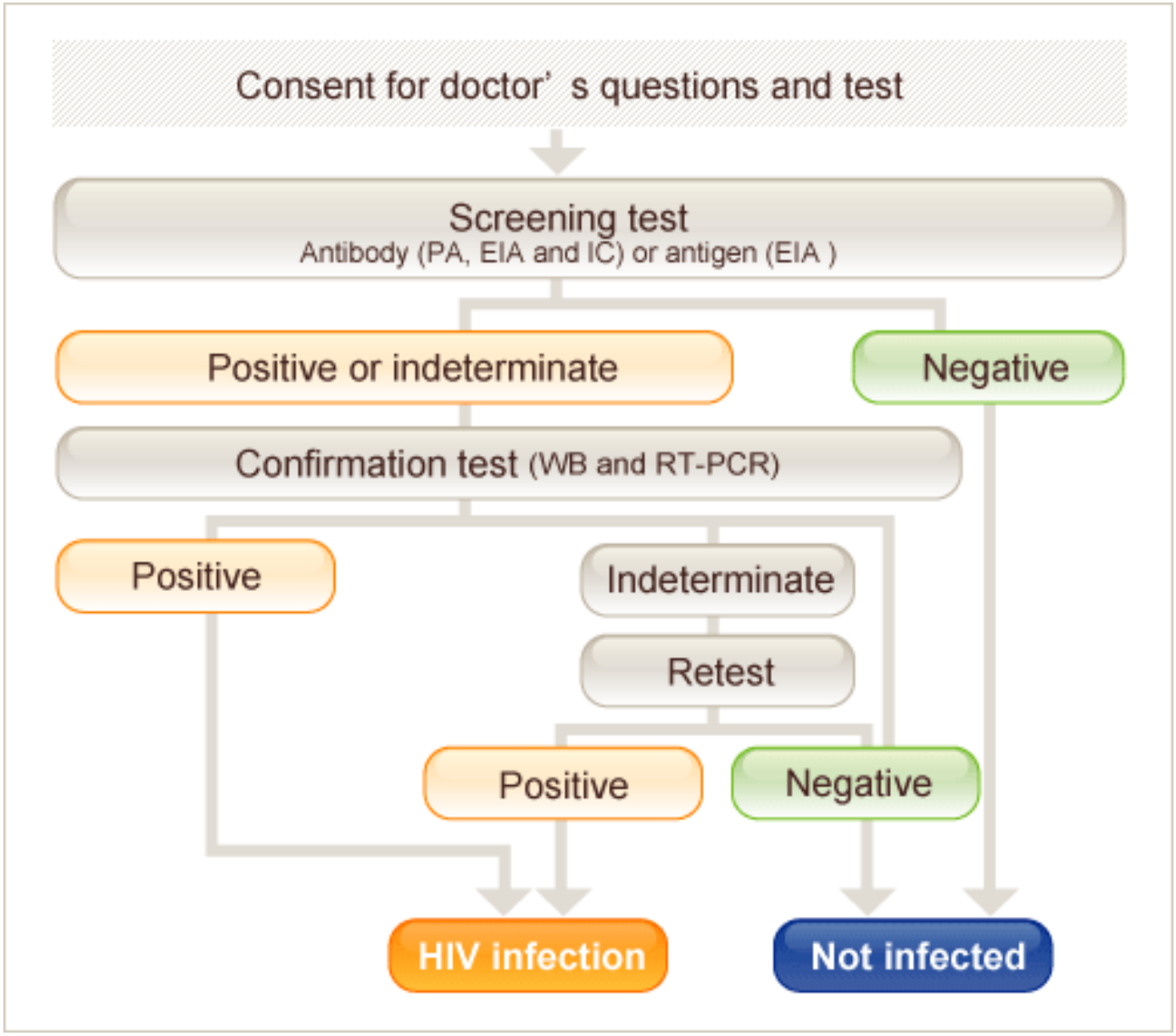
- The tests measure the amount of actively replicating HIV virus.
- Correlates with response to antiretroviral drugs
- Most helpful in diagnosis of HIV before seroconversion.



Uses of Absolute CD4 Lymphocyte counts :

- Most widely used predictor of HIV progression.
- Risk of progression to an AIDS defining illness, opportunistic infections or malignancy is high, when the counts drop below 200/ul

- Normal 800-1200 cells/ul
- >500; generally healthy
- 200-499; minor immune problems . Immune problems start when counts drop below 500 cells/µl
- <200; severe problems
- Estimation of percentages is useful. When the counts drops < 20% we have to watch for onset of opportunistic infections and malignancy (Ratio of CD4+ to CD8+ is reversed from 2:1 to 1:2) .


Consent for doctor’ s questions and test
Screening test
Antibody (PA, EIA and IC) or antigen (EIA)
Positive or indeterminate
Negative
Confirmation test (WB and RT-PCR)
Positive
Indeterminate
Retest
Positive
Negative
HIV infection
Not infected

- **Summary :**
- HIV is a RNA of retrovirus family .
- Long incubation period .
- Patients are infective all over their life .

- -Immune cells (CD4) infected with the virus .
- -AIDS patients are prone to secondary infections
- -confirmatory test are needed to confirm infection with HIV .

QUESTIONS…???

- **References**

1-Jawetz, Melnick, & Ad Elberg's (Medical Microbiology).

2- Warren Levinson review of Medical Microbiology and Immunology.

- **Immunologic Manifestations** • Early stage slight depression of CD4 count, few symptoms, temporary. • Window of up to 6 weeks before antibody is detected, by 6 months 95% positive. • During window p24 antigen present, acute viremia and antigenemia.
- **Immunologic Manifestations** • Antibodies produced to all major antigens. • First antibodies detected produced against gag proteins p24 and p55. • Followed by antibody to p51, p120 and gp41 • As disease progresses antibody levels decrease.
- **Immunologic Manifestations** • Immune abnormalities associated with increased viral replication. • Decrease in CD4 cells due to virus budding from cells, fusion of uninfected cells with virally infected cells and apoptosis. • B cells have decreased response to antigens possibly due to blockage of T cell/B cell interaction by binding of viral proteins to CD4 site. • CD8 cells initially increase and may remain elevated. • As HIV infection progresses, CD4 T cells drop resulting in